

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.37244	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	76.68216	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1450.44374	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.206322	0.206322	295.0254	0.569242	15.225

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.37244.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 76.68216.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1450.44374.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.188377.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.206322.
- Held-out runtime ms: 295.0254.
- Robustness dispersion across held-out structure families: 0.008095.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.369185, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.299109, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.203551, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.096879, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.062885, errors=[]

Judge Note

Phase-9 CVRP Solver-Evolution Suite Summary

Metric	Baseline Nearest Neighbor	Baseline Clarke-Wright Savings	Baseline Regret Insertion LS	Evolved Solver (Phase-9)
Feasibility Rate (heldout/train)	1.0 / 1.0	1.0 / 1.0	1.0 / 1.0	1.0 / 1.0
Heldout Objective Gap	0.2734	**0.0738 (best)**	0.4664	0.2063
Heldout Penalized Gap	0.2734	0.0738	0.4664	0.2063
Heldout Runtime (ms)	42.4	76.7	1450.4	295.0
Robustness Dispersion	0.0028	0.0078	0.0869	0.0081

Analysis

- ****Feasibility:**** All methods, including the evolved solver, maintain perfect feasibility (100%) on held-out instances.
- ****Objective Gap:**** The Clarke-Wright baseline has the lowest heldout gap (0.0738), outperforming the evolved solver (0.2063) and others.
- ****Runtime:**** Nearest Neighbor is fastest (~42ms), Clarke-Wright moderate (~77ms), evolved solver slower (~295ms), and Regret Insertion LS is much slower (~1450ms).
- ****Robustness:**** All methods show low robustness dispersion; nearest neighbor is the tightest, evolved solver close to Clarke-Wright.
- ****Novelty & Complexity:**** The evolved solver exhibits significant code novelty (0.57) and higher complexity (15.2), indicating exploration beyond baselines.

Conclusion

The evolved solver achieves full feasibility and competitive robustness but ****does not clearly beat the best fixed baseline (Clarke-Wright savings) in objective gap or runtime**** across held-out CVRP families. It offers a tradeoff of moderate gap and runtime with higher novelty and complexity but without a definitive performance advantage over the strongest fixed heuristic baseline.